Syllabus

Unit - 1 : Data Transmission Fundamentals

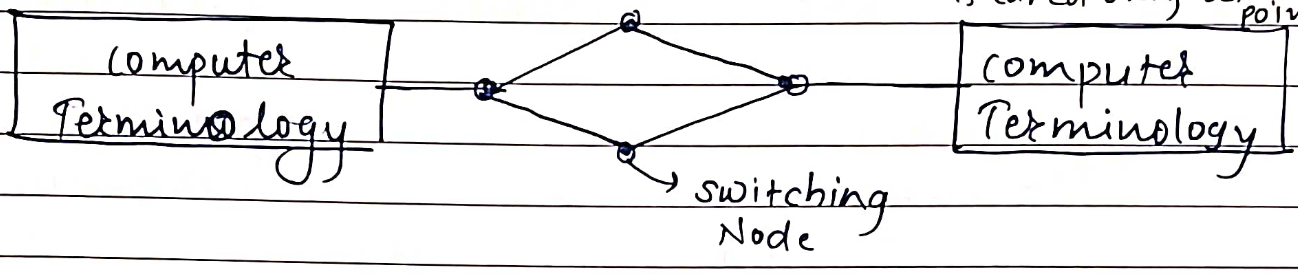
1. Data transmission concepts & terminology
2. Analog & digital data transmission
3. Transmission modes (simplex, half duplex, full duplex)
4. Transmission
5. Transmission Media : Guided (UTP, STP, Optical, coaxial) & Unguided (wireless {radio wave, micro wave, infrared})
6. Data Transmission : Parallel & Serial (Synchronous & asynchronous)
7. Analog & Digital signal Properties: Bandwidth, Bitrate, Baud rate, Data rate.
8. Connecting Devices : Hubs/ Repeaters, switches, Bridges, Routers.
9. Layred Architecture : OSI Model.

Unit 1

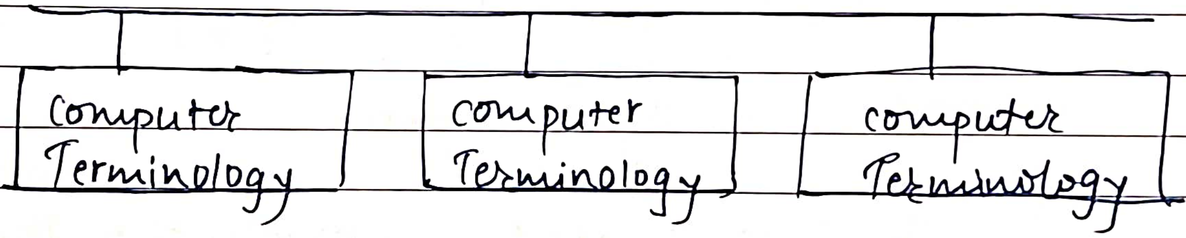
Data Transmission Fundamentals

* Data Transmission concepts & Terminologies.

1) Guided Media {Wired}
2) Unguided Media {Wireless}
3) Point to Point Transmission (Bandwidth is shared only betn 2 points)



4) Multipoint : (Bandwidth is shared betn multiple points)
Ex: LAN Network.



5) Direct Link :
Signal is transmitted directly to the transmitter using repeater.

1) Guided Media :
- Data is transmitted by electromagnetic wave through transmission media.
- The waves are guided along with physical path by guided media.
- Ex: Twisted pair cable, fibre optic cable, coaxial cable.

2) Unguided Media:

- There is no physical path to guide the electromagnetic wave.
- Ex:- satellite, Infrared, microwave, Wifi.

* Point To Point & Multipoint Configuration.

1)

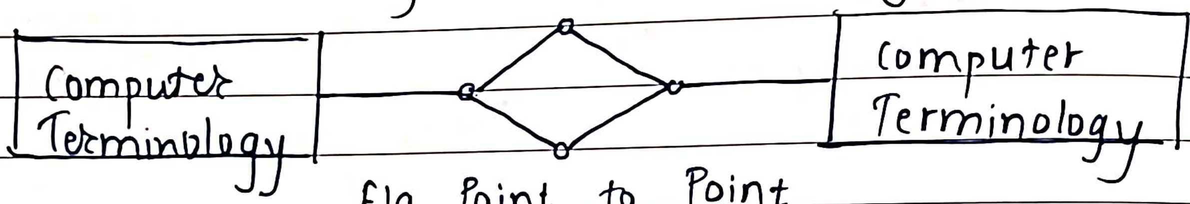


Fig. Point to Point

2)

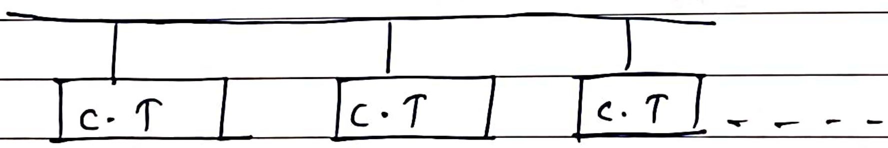


Fig. Multipoint

- In Point to point communication there is direct link between two terminals.
- Fig a. we observe that transmission medium is shared by only two communicating terminals.
- Fig. b. we observe that transmission medium is shared by many terminals.

* Direct Link:

- Signal is transmitted directly from transmitter to receiver.
- There is no intermediate device, so it is called direct link
- Amplifier & repeaters are used for increase the signal strength & remove the noise along direct link.

* Analog & Digital Transmission:

- Analog signal is basically continuous in time domain & amplitude.
- Digital signals are discrete in time & amplitude
- Most of the signals generated in nature are basically analog.
- Sometimes, signals are converted in digital form to make the storage & transmission easy.

1) Data: (Information) [Data carries some info] It can be analog or digital
2) signal
A signal is electric or electromagnetic representation of data.
Ex: Video signal is stored in CD, DVD to represent video.

3) Signaling:
Physical propagation of the signals through suitable medium is called signaling.
Ex: Radio waves through air.
Electrical signal through coaxial cable.

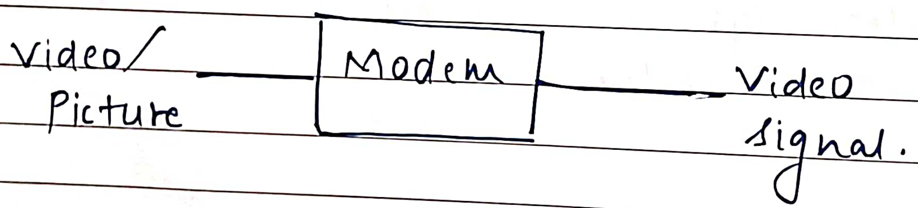
4) Transmission.
Communication & processing of signal combined is called transmission.

* Signals:

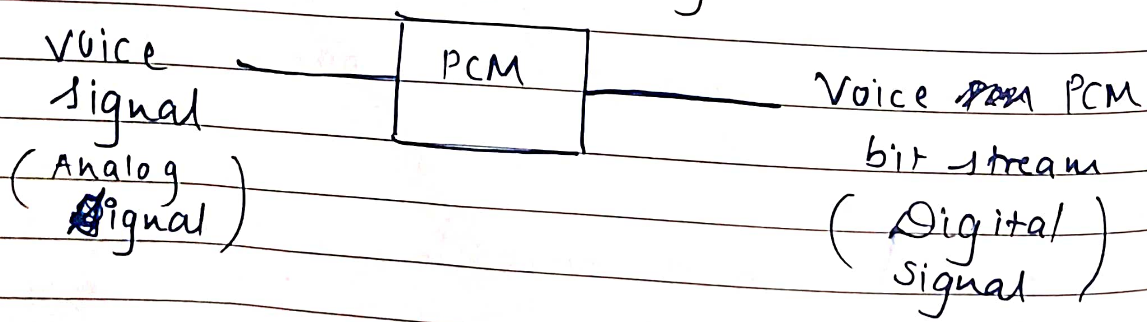
There are 4 types of signals:

1) Analog data to analog signal
2) Analog to Digital signal
3) Digital to digital. signal
4) Digital to Analog. signal.

1) Analog Data to Analog Signal :-



2) Analog Data to Digital signal



1) A to A :
- Voice is ~~an~~ in analog. voice signal is converted into electrical with the help of telephone.
- The electrical signal is analog in nature.

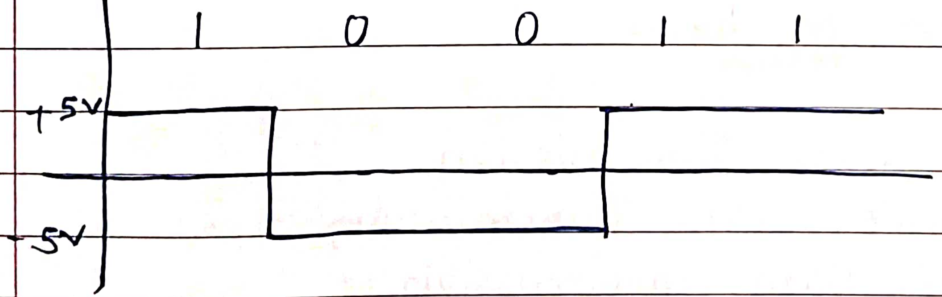
② Analog data to digital signal:

- Analog data voice, video, temperature ~~can be represented by~~ is applied to PCM.
- PCM converts this analog signal into voice PCM bit stream.
- Normally, PCM, DM, DPCM ~~&~~ can be used to convert analog data to digital.

3) Digital Data - Digital Signal.

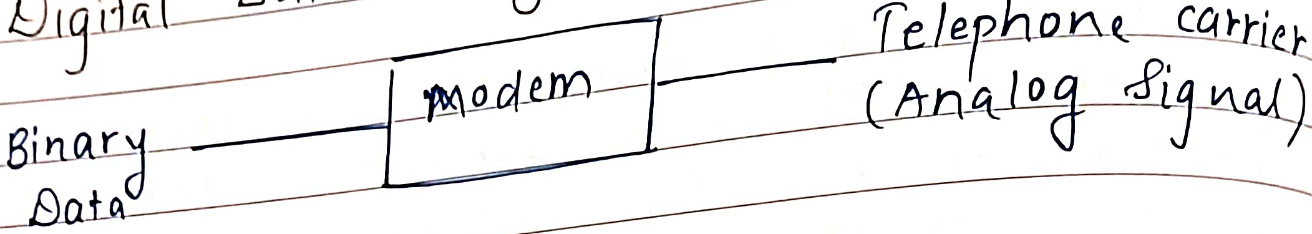
Binary Data (Digital Data) → | Digital Transmitter | → Encoded electric signal. (Digital signal)

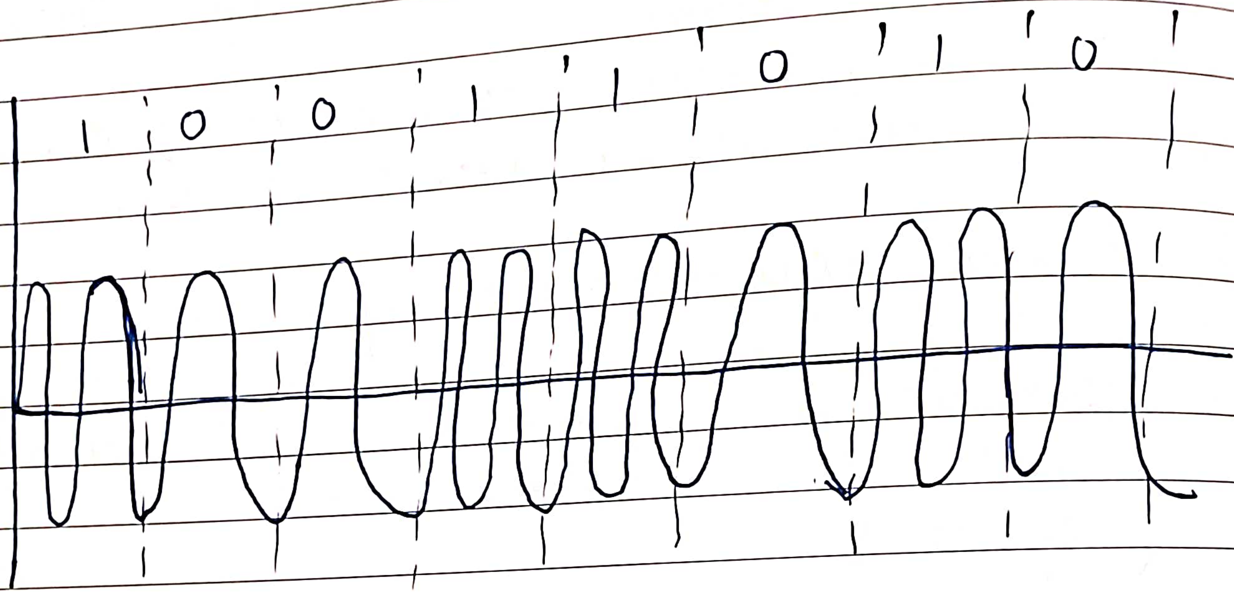
- Binary 1 can be represented by +5V & 0 can be represented by -5 V.
- This is called as Data encoding



- The transmitter encodes digital data into encoded digital signal.
- Ex: Printer.

4) Digital Data - Analog Signal.





- Binary data is applied to Modem.
- Modem converts digital data into analog telephone carrier

* Transmission Modes :

1) Simplex Data Transmission.
2) Half Duplex Data Transmission.
3) Full Duplex Data Transmission.

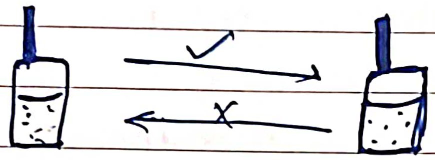
1) Simplex Data Transmission:

- Data can be transmitted in only 1 direction.
- Ex: Radio Transmission, from computer to printer, from mouse to computer.
- Data transmission is done only sender to receiver.

2) Half Duplex Data Transmission:

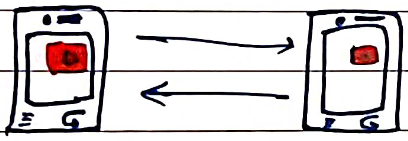
* ~~Data Transmis~~

- The data can be transmitted in both directions on a single carrier but not at same time.
- Ex: Walkie Talkie.



3) Full Duplex

- Data can be transmitted in both directions simultaneously
- Ex: Telephone, Mobile.



Mobile

* Transmission Media:

1) Guided Media.
2) Unguided Media.

Selection of Transmission medium depends upon following parameters:

1) Bandwidth: The max. data rate depends upon BW of transmission medium

2) connectivity: How the transmission medium can be connected in a networking.

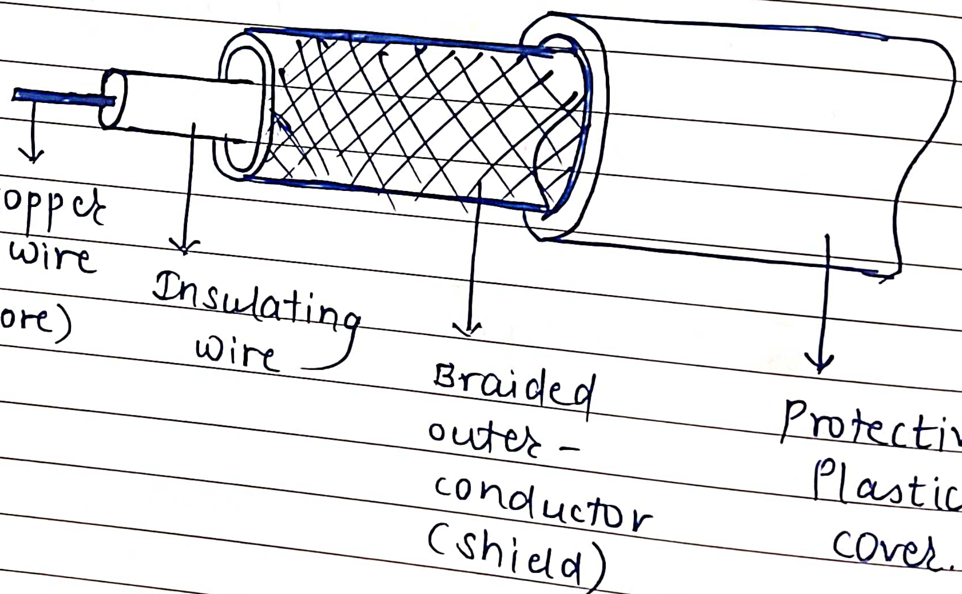
3) Geographic area: The parameter determines the geographic area covered by transmission media.

4) Security: The data security is important for selection of transmission media.

5) Cost: The cost should be minimum.

A) Guided Media / Wired Media

* Coaxial Cable



- It is widely used transmission medium in LAN.
- It has better shielding.
- It is used for long distance communication with higher speed.
- Coaxial cable has 2 types: 50 Ω & 75 Ω
- The 50 Ω cable is used for baseband digital transmission.
- The 75 Ω cable is used for broadband digital transmission.

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* Construction :

- coaxial cable consist of copper wire as a core. which is surrounded by insulating material.
- The insulator is covered by braided outer conductor (also called as shield)
- Shield is covered by protecting plastic cover
- This construction is suitable for high band-width & excellent noise immunity.
- The losses in coaxial cable are very low.

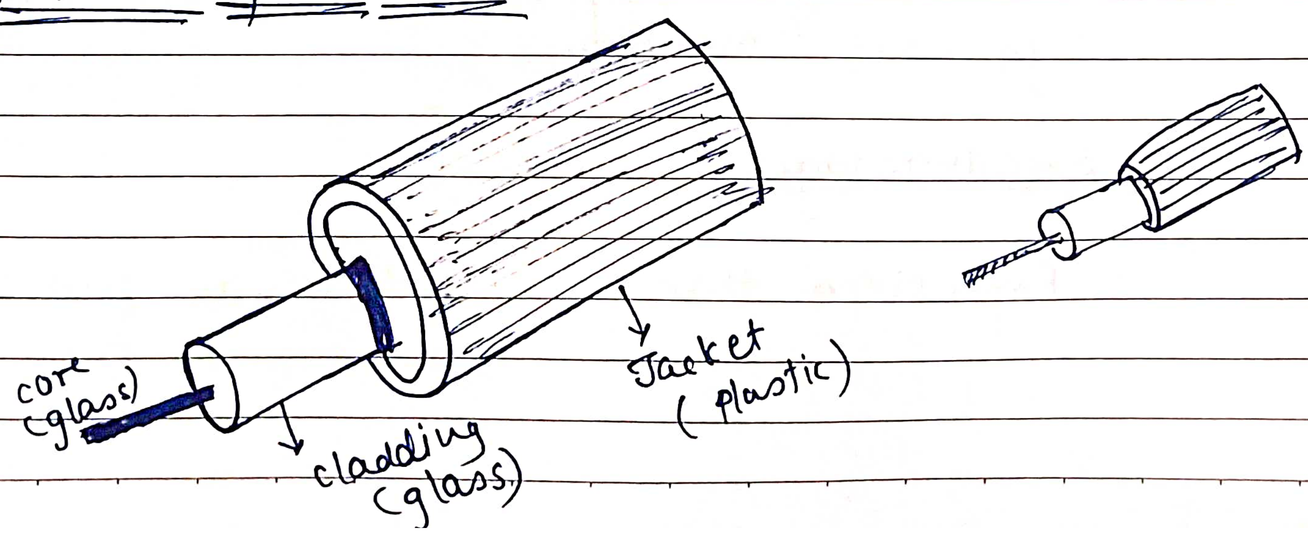
* Advantages:

- Easy to install.
- Not expensive. (than fibre optic cable)
- Less Dataloss.

* Disadvantages:

- Cable gets easily damaged.
- High cost as compared to twisted pair cable.

2) Fibre Optic cable:



- Signals are carried by light.
- There is no electric current & electric voltage passed through fibre optic cable.
- Light source & detectors are used to convert light into electrical signal.
- Fibre optic cable offers minimum losses because light is used as a signal.
- Due to light signal, bandwidth of FOC is very large.
- It is used for long distance communication.
- It has very high data rates (gbps)

* Construction:

- Core is surrounded by cladding
- Core & cladding are glass
- Cladding is surrounded by protective plastic Jacket.

* Advantages:

- High transmission speed.
- FOC cable offers minimum losses.
- It is used for long distance communication.
- High noise immunity.

* Disadvantages:

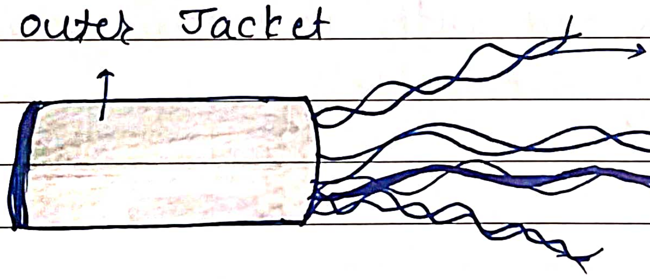
- Expensive than coaxial & twisted pair cable.

3) Twisted Pair cable:

- This is the cheapest transmission medium & commonly used.
- BW. of twisted pair cable is less.
- Used for short distance communication.
- The TPC consist of 2 ~~a~~ insulated wires in spiral form

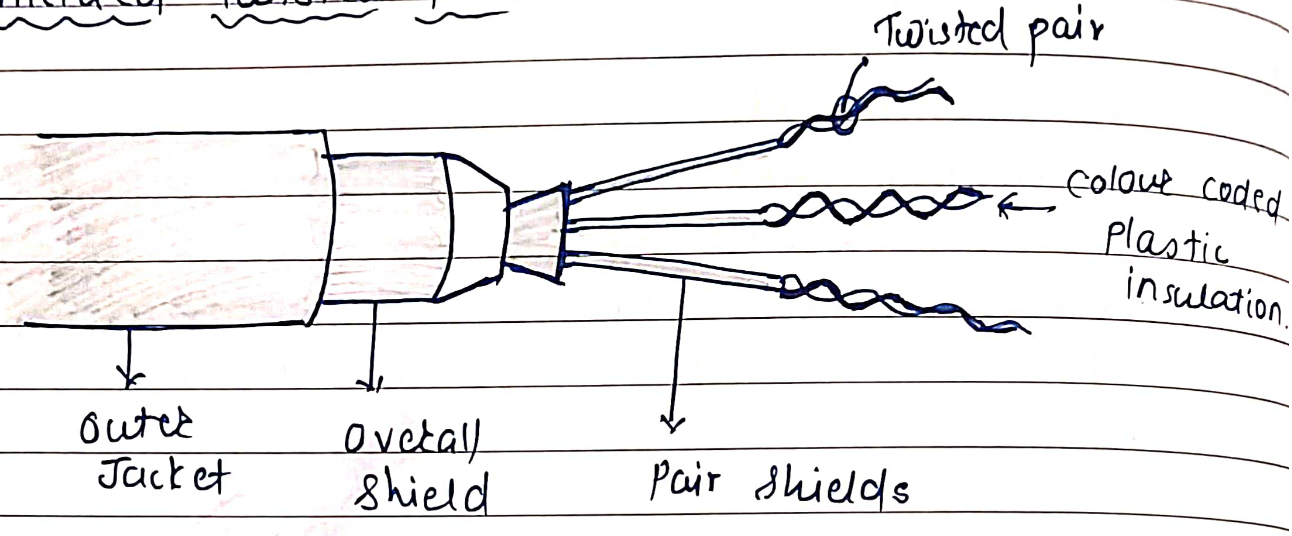
Twisted pair cable consist of 2 types:
 i. Unshielded Twisted pair cable. (UTP)
 ii. shielded Twisted Pair Cable (STP)

i. Unshielded Twisted Pair Cable: (UTP):



- The UTP wire is most inexpensive transmission media.
- It is very easy to use & ~~at~~ simple to install.
- The UTP cables can easily be affected due to noise.

ii) Shielded Twisted Pair (STP)



- To overcome the effect of noise, Shielded Twisted Pair cable is used.
- The metallic braid / shielding is used around twisted pair cable, which reduces the effect of interference.
- The wire is covered with protective plastic covering.
- Data transfer rate of STP cable is very low.

* Advantages:

- Easy to install
- Less expensive.

* Disadvantages:

- Used for short distance communication.
- Easily affected due to noises

B) Unguided Media / Wireless Transmission.

1. satellite
2. Terestrial
3. Infrared Transmission.
4. Broadcast radio.

1) Satellite

a)

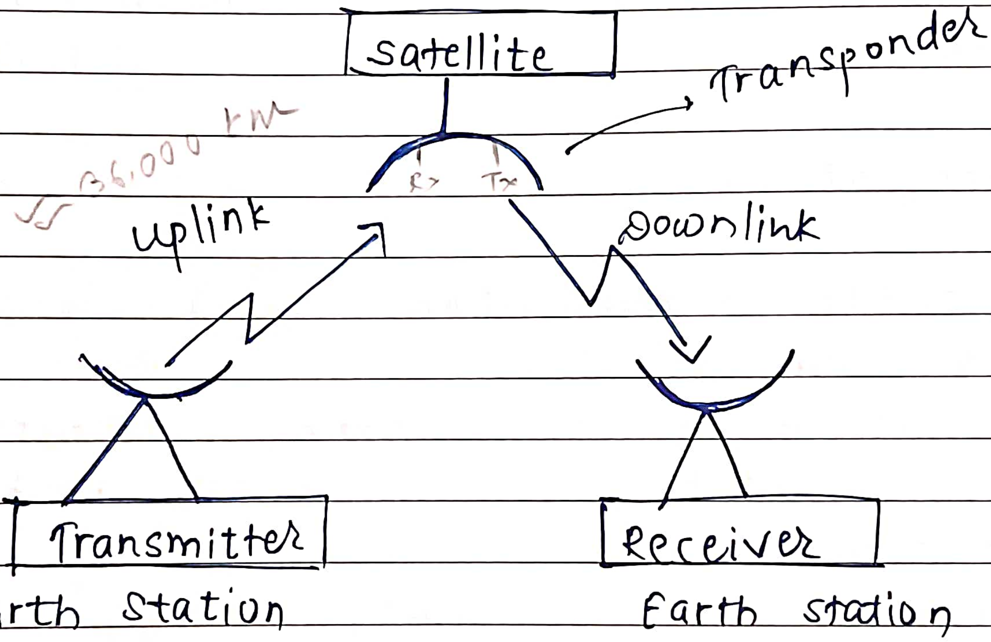


fig. Point to point link

b)

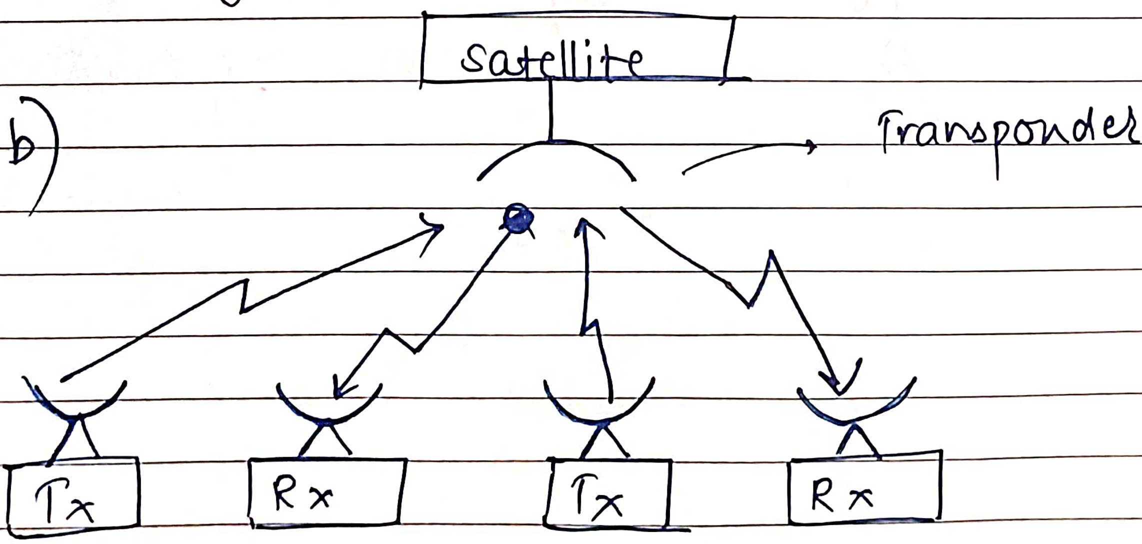
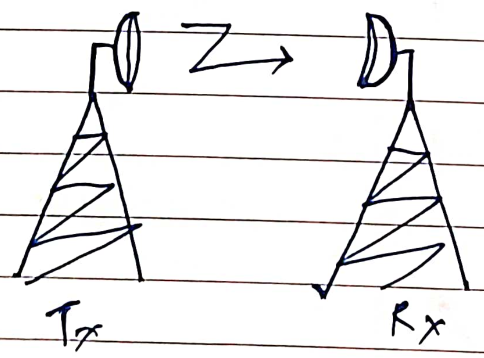


fig. Multipoint or link via Satellite

- Satellite are widely used for repeaters.
- Satellite consist of receiving antenna & receiver circuit
- The transmitted signal on coupling frequency is received by satellite. The signal is filtered, amplif and re-transmitted on down link frequency.
- The transmitter & receiver of satellite consist of one channel called as transponder.
- Transponder has common transmitting & receivi antenna.

* Applications

1. Satellite communication.
2. Television Distribution.
3. VSAT are used for providing buisness network.
4. For weather forcasting, military mobile radio transmission.
5. Line of site communication.



2) Infrared Transmission:

- Infrared light is used for communication.
- It is strictly line of sight communication.
- Infrared waves cannot penetrate the walls, hence, this type of transmission mainly used for security applications.
- Noise interference is less

* Applications:

1. In remote control of home appliances

3) Broadcast Radio:

- It uses the frequency range from 30 MHz to 1 GHz. for transmission.
- This frequency is called as Radio Waves.
- Transmission is mainly omini-directional. Hence, it does not require directional antenna.
- The distance is covered by:

$$d = 7.14 \sqrt{kh}$$

* Application

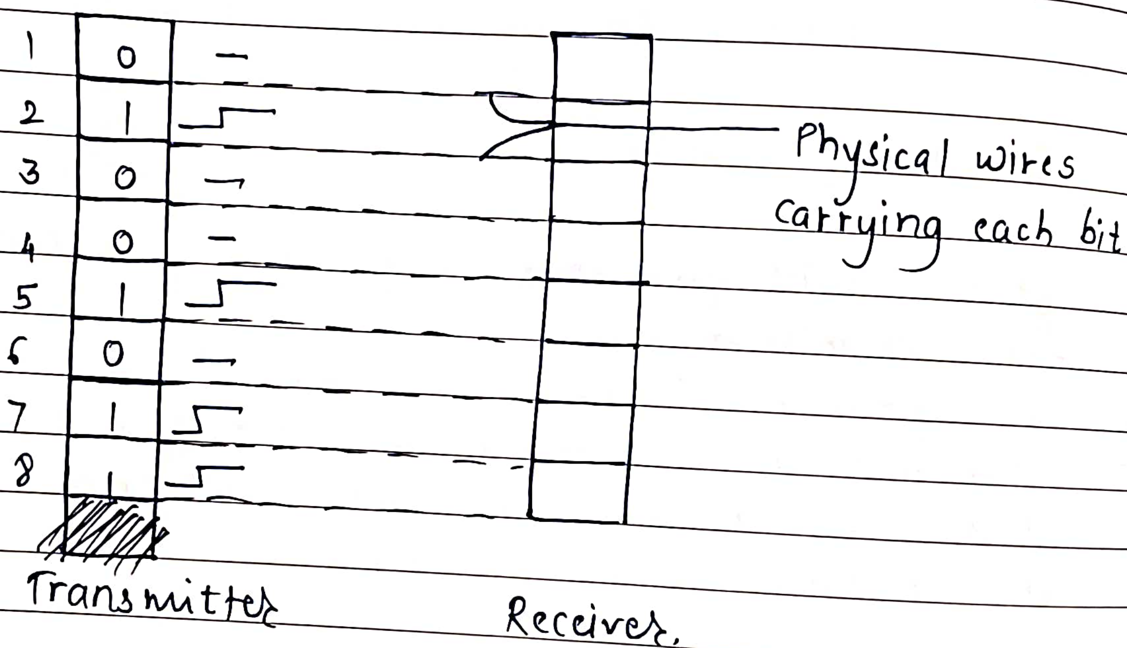
1. Broadcasting of FM radio,
2. Broadcasting of UHF & VHF radio.

* Data Transmission:

Data Transmission has 4 types

1) Parallel Data Transmission.
2) Serial Data Transmission.
3) Synchronous Data Transmission.
4) Asynchronous Data Transmission.

1) Parallel Data Transmission:



- The no. we have is 8 binary digit.
- When the data transmission is required, all the bits of a byte are transmitted simultaneously on seperate wires.
- This is called parallel transmission.
- From fig., there are 8 wires for 8 bits. Parallel transmission is used for short distance communication.

- In parallel transmission, at a time 8 bits data is transmitted simultaneously because, ~~it~~ each bit uses seperate wire for transmission.

Ex:- Parallel transmission takes place when printer is connected to parallel port of connector.

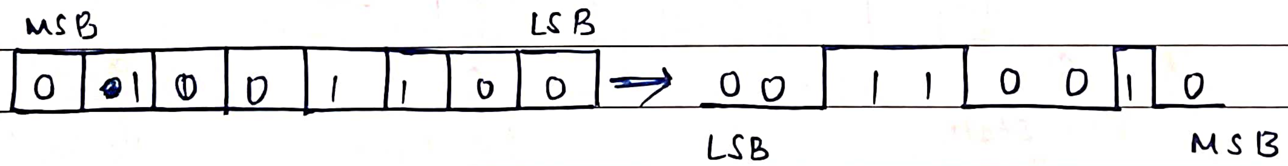
Advantages:

- Fast data transfer rate.
- Complexity is less

Disadvantages:

- Cost is more.
- Not suitable for long distance transmission.

2) Serial Transmission:



- Bits of byte are transmitted serially one after another.
- The bits stored in register shifted towards LSB ~~one~~ at every clock pulse.
- When the clock pulse is received, the bit at LSB position is transmitted on line.
- Serial communication take place on a ~~serial~~ single wire between transmitter & receiver.

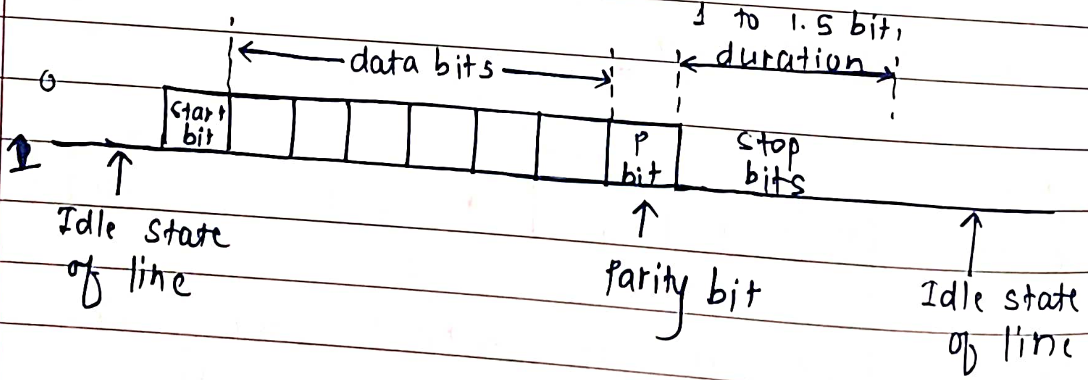
* Advantages:
- Reduced no. of wires for connecting 2 system
- Most economical way of transmission.

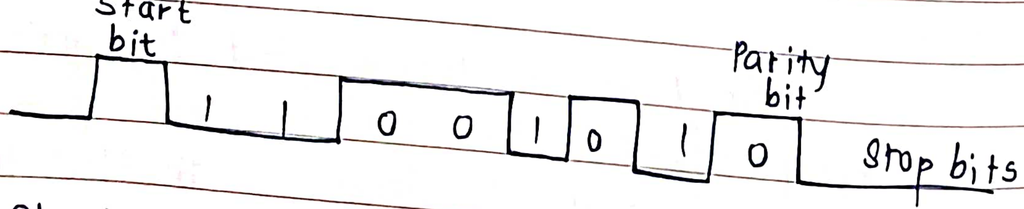
* Disadvantages:
- Serial transmission is slow.
- Complexity is more.

* Application:

- LAN uses serial transmission.
- Long distance communication.

3) Asynchronous Data Transmission.

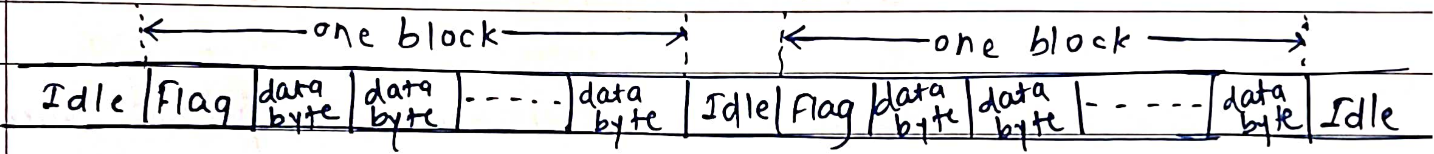




- Short string of binary data is transmitted. These strings are upto 8 to 10 bits. Start & Stop bit are used for proper synchronization bet^n transmitter & receiver.

- Fig. shows, it consists of 7 characters.
- 1 start bit is set; it idicates begining of character.
- After start bit, 7 bits of 1 character / data bits are transmitted.
- 1 parity bit is added after data bits.
- After parity bit, 1 to 1.5 bit duration is transmitted.

4) **Synchronous Data Transmission:**



- Transmitter & receiver operate on common clock signal.
- There are no start and stop bit.
- Bytes are transmitted as block or frame.
- Synchronous transmission is called as coherant transmission.
- It is complex and number of errors are reduced/ minimized.

* <u>Hub, switch, Router</u>:

* <u>OSI Model</u> (Open System Interface)

	Endsystem	Subsystem			Endsystem
Layer 7	Application layer	Protocol			Application layer
Layer 6	Presentation layer	Protocol			Presentation layer
Layer 5	Session Layer	Protocol			session layer
Layer 4	Transport layer	Protocol			Transport layer
Layer 3	Network layer	↔ Network layer		Network layer ↔	Network layer
Layer 2	Data link layer	↔ Data link layer		Data link layer ↔	Data link layer
Layer 1	Physical layer	↔ Physical link		Physical link ↔	Physical layer
	Transmission media				Transmission Media

- Data communication system should follow some compatible of world wide standards.
  These standards fit into framework establish by ISO (India standard Organisation)
  This framework is called as model for open system Interconnection. (OSI)

- The lowest layer is called 'physical layer' which specifies conversion of bits into electrical signal, for connection of network.

- i) Application layer:
  - Application layer is the highest layer in OSI model
  - It provides services to user
  - Ex: Login, password checking, file transfer are performed by application layer

2) Present
- The 2 communicating system may have different local syntax / code and data format differences.
- These differences ~~are resolved~~ in format are resolved by presentation layer.
- Ex: 1 end of the system transfers the file in ASCII code format & other end of the system need this file but it uses EBCDIC format. Then translation from ASCII to ECEDIC format will be done in the presentation layer.

3) Session layer:
- Management & synchronisation of dialogue (conversion) betn 2 different applications is done in session layer.
- User establich ~~session to s~~ system to system.
- Ex: Login on off, user identification, billing & session management are controlled by this layer

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* Analog & Digital Signal Properties.

1) Bandwidth:
Bandwidth is the difference betⁿ highest frequency & lowest frequency of signal.

2) Bit rate:
- The rate at which individual bits of signal are transmitted is called as Bit rate.
- Unit: bps (bit per second)
- It is used for determining BW & channel capacity.

3) Baud Rate:
- The rate at which changes occur in a system over given period of time.
- Baud is a unit of signaling speed &/ modulation rate.

4) Data Rate:
- Rate at which data is transmitted / needs to be transmitted is called data rate.
- Unit → Bits per sec, samples per sec